Project Documentation: LDU Factorization Using Elimination Matrices

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1. ProjectOverview:

Course : Introduction to Computer Science
Topic : LDU Factorization Calculation
Repository : Github

Objective:

This website allows the users to input a matrix of any dimension and calculate its LDU (Lower Diagonal Upper) factorization. The website will display the factorization process step by step, Showing the L, D and U matrices that result from the decomposition

Purpose:

This document provides an explanation of the LDU Factorization method, step-by-step, and applies it to an example. This documentation is prepared by a team of five members.

2.Team Members and Contributions:

Name	Role	Contribution
Pranjal Dubey	Algorithm Design & Optimization and Project Manager.	• **Styling Contribution:** Enhanced documentation clarity through styling improvements. • **Web Development**: Designed web pages using HTML. • **Bug Identification and Optimization:** Identified potential bugs, performance bottlenecks, and areas for improvement in the factorization process.
Shalini Priya	Project Manager and Frontend developer.	• **Documentation Management:**

		Oversaw the documentation process to ensure all project aspects were thoroughly documented. ● **Resource Allocation and Risk Management:** Ensured efficient allocation of resources and managed potential risks.
Manoj Dewda	Documentation part (Writing & Formatting) and Frontend developer.	● **Documentation Management:** Oversaw the documentation process to ensure all project aspects were thoroughly documented. ● **Resource Allocation and Risk Management:** Ensured efficient allocation of resources and managed potential risks.
Shivam Sindhu	Documentation part (Writing & Formatting) and Frontend developer.	● **Documentation Management:** Oversaw the documentation process to ensure all project aspects were thoroughly documented. ● **Resource Allocation and Risk Management:** Ensured efficient allocation of resources and

		managed potential risks.
Prem Kumar	QualityAssurance (Testing & Debugging).	● **Thorough Testing and Debugging:** Conducted extensive testing and debugging of the LDU factorization code to ensure accuracy and efficiency of the algorithm.

4. Project Workflow and Tools:

- **Tools Used:**

1. **Version Control**: GitHub.
2. **Communication:** Google Meet, and Whatsapp.
3. **Languages:** HTML, CSS.

- **Workflow:**

1. We adopted the GitHub flow for collaboration, beginning with separate branches for each feature and merging them after a thorough review.

5. Development Process

1. Phase 1 - Planning:

❖ We all members of our group discussed how to execute the things.

❖ Established the project scope and assigned tasks to the team members.

❖ We created a GitHub repository and set up basic files and folders.

2. Phase 2- Implementation:

❖ “Then, the UI was designed with the input fields for matrix dimensions and elements.

❖ Tests were conducted on various matrices to verify the accuracy of the results."

3. Phase 3- Review and Testing:

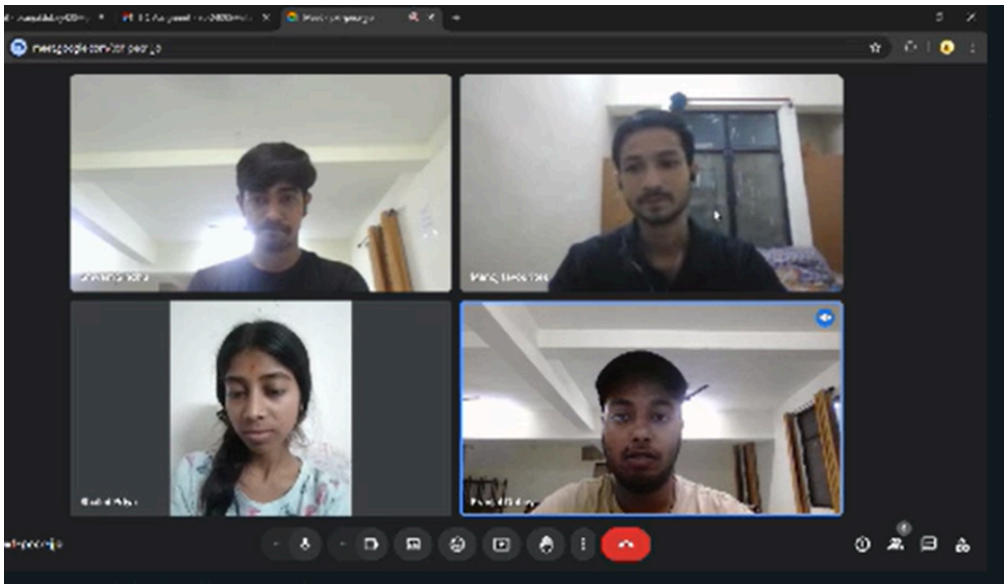
❖ Meetings were held to evaluate progress.

❖ Bugs and areas for improvement were identified and addressed collaboratively by the Team.

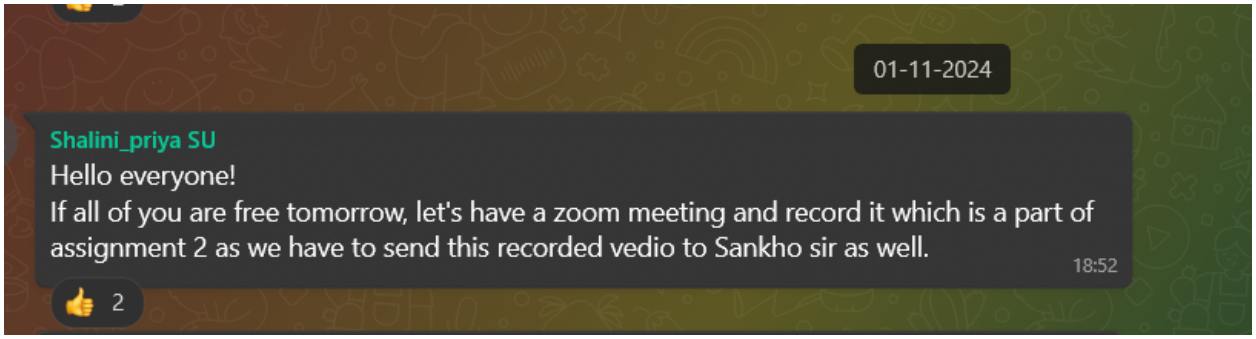
❖ Ongoing discussions were hello to simplify and enhance the user-friendliness of the project.

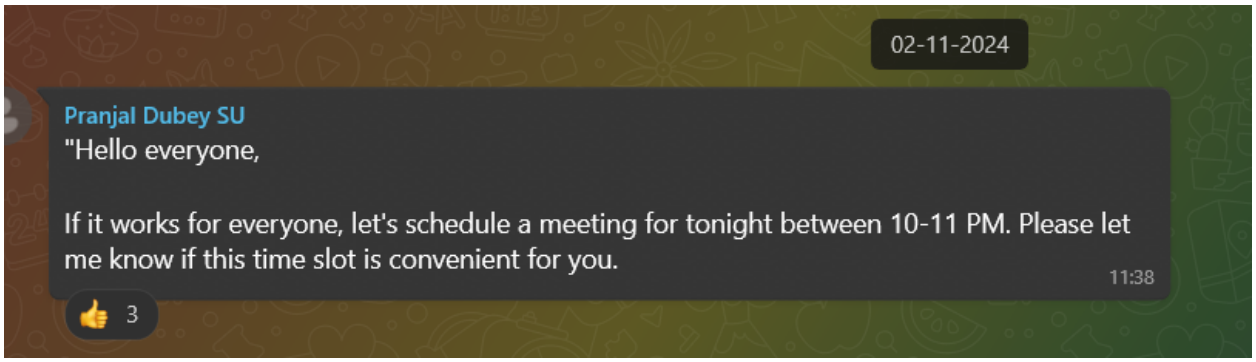
6. Communication And Discussion About Project:

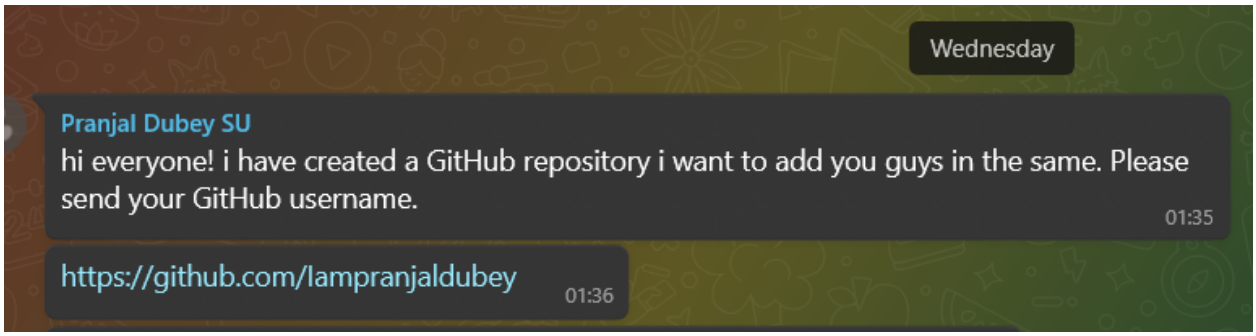
Google Meeting Screenshots:



Whatsapp screenshots:







Steps for LDU Factorization:

LDU factorization is a matrix decomposition method where a square matrix AAA is decomposed into three components: a lower triangular matrix LLL, a diagonal matrix DDD, and an upper triangular matrix UUU, such that:

A=L⋅D⋅UA = UA=L⋅D⋅U

Here are the general steps to perform LDU factorization:

1. Input Matrix

Start with a square matrix AAA of size n×nn \times nn×n.

2. Decompose into LU Form

First, decompose AAA into an LU factorization. LU factorization breaks down the matrix into:

- LLL (lower triangular matrix)
- UUU (upper triangular matrix)

Steps for LU Factorization:

- Use Gaussian elimination or other methods (like Doolittle or Crout's method) to perform LU decomposition.
- LLL contains ones on the diagonal (in Doolittle's method) or may contain arbitrary values on the diagonal (in Crout's method).
- UUU contains the upper triangular portion of the matrix.

3. Extract the Diagonal Matrix DDD

- After obtaining LLL and UUU, create the diagonal matrix DDD by extracting the diagonal elements of UUU.
- The diagonal elements of UUU become the diagonal elements of DDD, while the off-diagonal elements in UUU are set to zero.

4. Adjust the Matrices

- Once DDD is extracted, adjust LLL and UUU accordingly:
    - LLL will now contain the lower triangular elements, with ones on the diagonal.
    - UUU will be adjusted to exclude the diagonal, which is now part of DDD.

5. Final LDU Decomposition

After adjusting the matrices, the decomposition is complete:

A=L⋅D⋅UA = L \cdot D \cdot UA=L⋅D⋅U

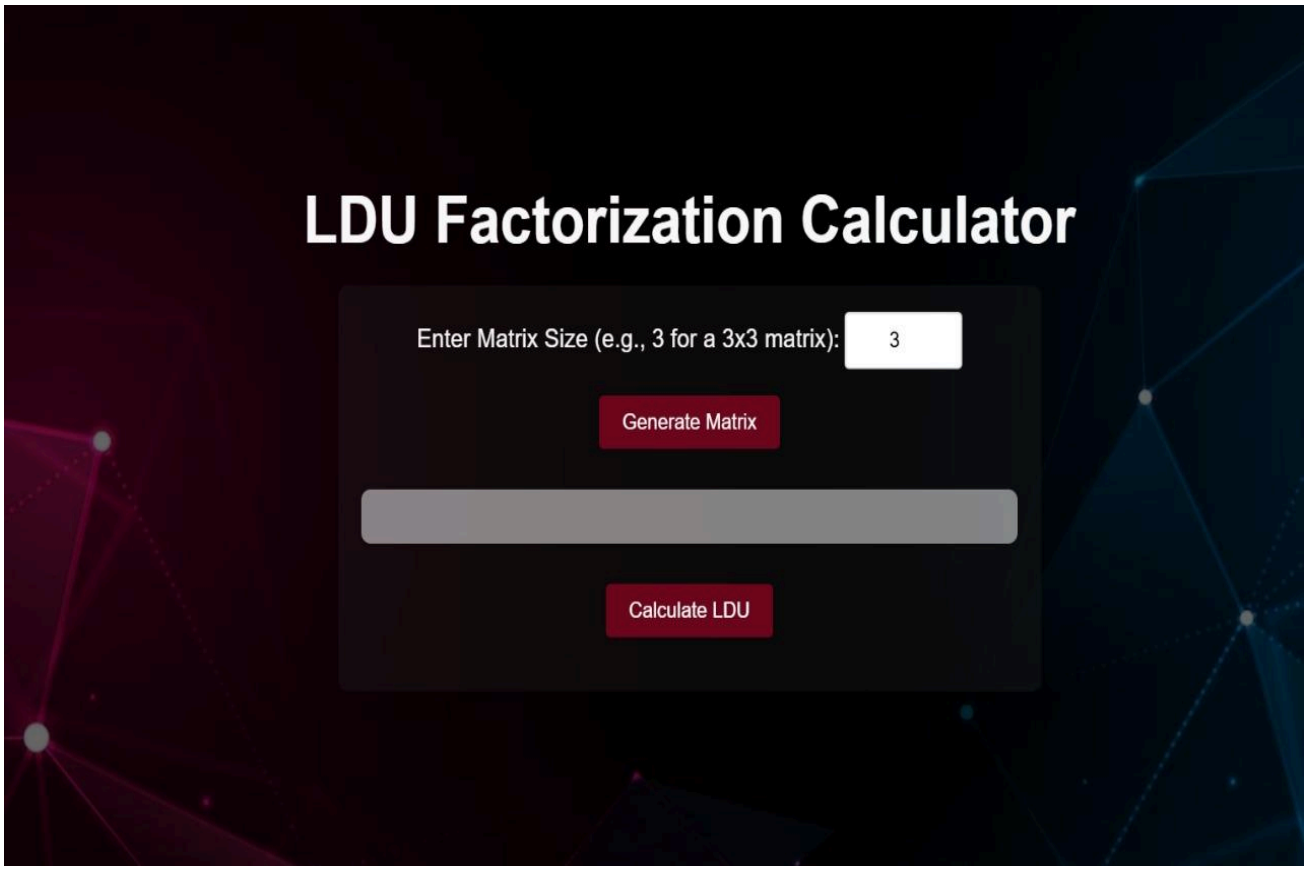
- LLL: Lower triangular matrix with ones on the diagonal.
- DDD: Diagonal matrix.
- UUU: Upper triangular matrix without the diagonal elements.

Project Features:

● **Matrix Input Interface:** Provides an interactive way for users to specify the dimensions and elements of the matrix
● **LDU Decomposition:** Performs a detailed, step-by-step breakdown of the input matrix into its L, D, and U components.

● **Scalable:** Handles both small and large matrices, suitable for high-performance computing.
● **Customizable Structure:** Allows flexible partitioning of the matrix into four parts based on problem needs.

References:

● Textbooks and online resources on LDU factorization.
● GitHub repository and documentation standards.

How our calculator look like: